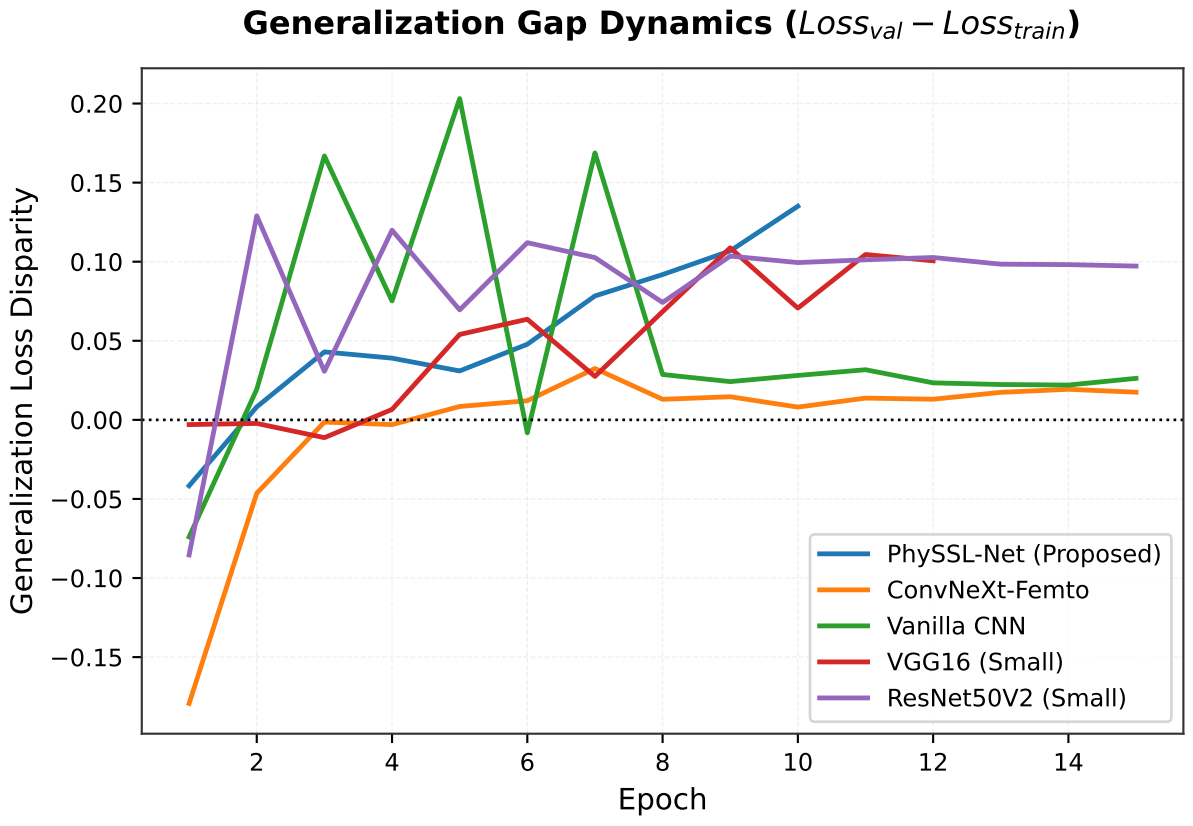
Generalization Gap Dynamics ($Loss_{val} - Loss_{train}$)
Generalization Loss Disparity
Epoch
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
2
4
6
8
10
12
14
PhySSL-Net (Proposed)
ConvNeXt-Femto
Vanilla CNN
VGG16 (Small)
ResNet50V2 (Small)